

Standing Push Recovery Based on LIPM Dynamics Control for Biped Humanoid Robot

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Abstract—This paper presents a control method based on LIPM (Linear Inverted Pendulum Model) dynamics for the balance control of biped robot. By setting CoM (Center of Mass), CP (Capture Point), and ZMP (Zero-Moment Point) as the state of LIPM, we use state feedback of linear system to control all of them synchronously. Considering that the robot control system is digital, analysis and controller design are based on discrete system. Simulation results verify the effectiveness of the proposed method. It is also evaluated in the experiments of the robot WALKER standing push recovery.

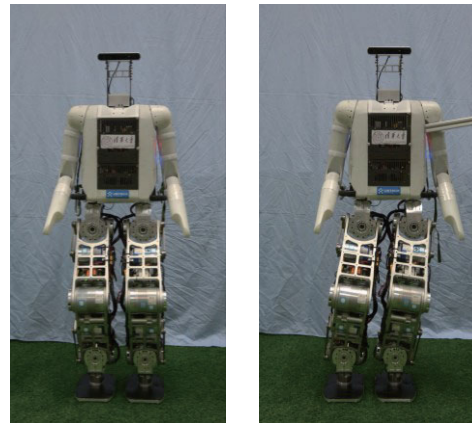
I. INTRODUCTION

The stable locomotion of biped robots has many challenges because of multi degree of freedom, complicated dynamics and varying contact situation. As a consequence, the problem is usually divided into motion trajectory generation and balance control with simplified models, which hold only some essential aspects of the multi-body dynamics.

Vukobratovic introduced the Zero-Moment Point (ZMP), which is a concise representation of the vertical force distribution between the foot and the ground [1]. And it can be a criterion for the balance of a biped robot. Many researchers relate the ZMP to the Center of Mass (CoM) and prove that it is useful for trajectory generation and stabilization.

Kajita et al. proposed the linear inverted pendulum model (LIPM) that is simple enough to generate walking pattern easily [2] and also used preview controller by future reference ZMP [3]. The model has been widely popularized and improved with other characteristics. Tsuji et al. presented Virtual Supporting Point (VSP) to be the pivot of the LIPM so that the stability margin can be broadened [4]. Shibuya et al. used the LIPM with VSP to generate the CoM trajectory and also got a continuous ZMP trajectory [5]. Pratt et al. proposed the Capture Point (CP) of LIPM [6], also called ‘Extrapolated Center of Mass’ by Hof in [7]. The CP is the special point of LIPM which lead the robot to a complete rest. While Pratt et al. discussed its significance for biped push recovery, Hof used it for designing step-to-step walking controller. Englsberger et al. extended CP to 3D, called the Divergent Component of Motion (DCM) for motion planning [8].

Also, there are various strategies of balance control around a given pattern. Kajita et al. proposed linear inverted pendulum tracking control with a first order model of the ZMP lag [9]. Choi et al. presented stability analysis by Lyapunov method for a ZMP based balancing controller [10]. Using the concept of CP, two approaches for CP control for balance control were put forward by Englsberger et al. [11]. Takenaka et al. had also focused on the DCM which is



(a) WALKER when standing (b) Reaction when pushed

Fig. 1. Standing push recovery of robot WALKER

equivalent to the CP [12]. The landing position is modified to prevent losing the balance according to the divergent component as a result of the model ZMP control which accelerates the upper body. Morisawa et al. improved a balance control based on the CP control with the CP integration [13].

This paper uses VSP and takes the CoM, CP and ZMP as three state of LIPM. Then we propose the state feedback control method of discrete LIPM system and prove the controllability of it. Simulation results can verify the effectiveness when there are some deviations between LIPM and the real system. Our experimental platform is the Walker shown in Fig. 1(a), which is a position-controlled biped robot. When the robot is standing and pushed, the controller will change the CoM to resist the disturbance as shown in Fig. 1(b). In Section II, we introduce the LIPM with VSP in the form of the discrete system, then derive the equation of the system by state space description. Controllability proof and pole assignment are also clearly presented. In Section III, we show the application details to the actual robot of this method. In Section IV, simulation and experimental results are presented. In Section V, we make a conclusion and discuss the further application of the proposed method. In the appendix, the details of the lengthy formula are shown.

II. LIPM COM-CP-ZMP CONTROL

A. LIPM, VSP and CP

We use the LIPM to generate the CoM trajectory of the

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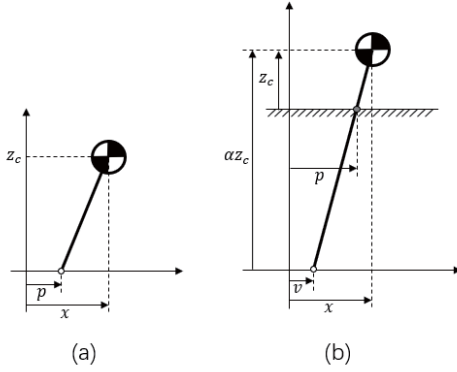


Fig. 2. (a) Linear Inverted Pendulum Model
(b) LIPM with Virtual Support Point

robot. The LIPM shown in Fig.2(a) has the dynamics of

$$\ddot{x} = \omega^2(x - p), \quad (1)$$

where ω is a natural frequency of the LIPM defined as

$$\omega \equiv \sqrt{\frac{g}{z_c}}.$$

The natural frequency ω is composed of the CoM height z_c and gravity constant g .

Generally speaking, the pivot of LIPM is on the ground plane, and for biped robot it is Zero-Moment Point. However, when it is on a virtual plane under the ground as shown in Fig. 2(b), we can make the ZMP trajectory continuous and fit for actual foot of the robot. When the pendulum height is α times as long as the original height, the dynamics equation is written as

$$\ddot{x} = \frac{\omega^2}{\alpha}(x - v) \quad (2)$$

And then the ZMP can be expressed as a state of the LIPM, related to the position of CoM and the VSP.

$$p = \frac{\alpha - 1}{\alpha}x + \frac{1}{\alpha}v \quad (3)$$

In addition, Capture Point is also an important characteristic of LIPM, corresponding a state of $(x, \dot{x})$.

$$\xi = x + \frac{\sqrt{\alpha}}{\omega}\dot{x} \quad (4)$$

And ξ has the dynamics of

$$\dot{\xi} = \frac{\omega}{\sqrt{\alpha}}(\xi - v) \quad (5)$$

B. CoM-CP-ZMP Control Method

Based on the above considerations, we take $\mathbf{x} = (x, \xi, p)$ as the state of LIPM with VSP which can represent it completely. When we generate the CoM trajectory using LIPM, the planning state at each time can be a reference of control. Because of using digital system and timing loop of control, we should write down the equation of state transition from the last time n to the next time $n + 1$ on the cycle time T .

According to the analytic solution of (2), we find

$$x_{n+1} = (x_n - v_n) \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \frac{\sqrt{\alpha}}{\omega}\dot{x}_n \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + v_n \quad (6)$$

$$\dot{x}_{n+1} = \frac{\omega}{\sqrt{\alpha}}(x_n - v_n) \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \dot{x}_n \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \quad (7)$$

Corresponding (5), we obtain

$$\xi_{n+1} = \frac{\omega}{\sqrt{\alpha}}(\xi_n - v_n) \quad (8)$$

Considering that the real ZMP lags behind the ideal in our model due to the mechanical compliance and the control, we use the first order dynamics to model this effect.

$$p_n^d = \frac{\alpha - 1}{\alpha}x_n + \frac{1}{\alpha}v_n \quad (9)$$

$$p_{n+1} = -g_p p_n + g_p p_n^d \quad (10)$$

The p_n^d is ideal ZMP of the LIPM in (9) corresponding (3), and the first order dynamics is in (10), where the g_p should be a parameter related to the actual robot's mechanical characteristic.

Using all of (6) – (10), we can obtain the state transition equation from (x_n, ξ_n, p_n) to $(x_{n+1}, \xi_{n+1}, p_{n+1})$ with a form of

$$x_{n+1} = f_1(x_n, \xi_n, v_n) \quad (11)$$

$$\xi_{n+1} = f_2(x_n, \xi_n, v_n) \quad (12)$$

$$p_{n+1} = f_3(x_n, \xi_n, v_n) \quad (13)$$

Because of the length of these equations, the specific content is attached in the appendix. But it is also enough for the later analysis.

The control objective is the reference state, so we should find a control input to eliminate the influence of state deviation. The complete differential for these three states can be written with the form of

$$\Delta x_{n+1} = \frac{\partial f_1}{\partial x_n} \Delta x_n + \frac{\partial f_1}{\partial \xi_n} \Delta \xi_n + \frac{\partial f_1}{\partial v_n} \Delta v_n \quad (14)$$

$$\Delta \xi_{n+1} = \frac{\partial f_2}{\partial x_n} \Delta x_n + \frac{\partial f_2}{\partial \xi_n} \Delta \xi_n + \frac{\partial f_2}{\partial v_n} \Delta v_n \quad (15)$$

$$\Delta p_{n+1} = \frac{\partial f_3}{\partial x_n} \Delta x_n + \frac{\partial f_3}{\partial \xi_n} \Delta \xi_n + \frac{\partial f_3}{\partial v_n} \Delta v_n \quad (16)$$

We can use state space description of discrete system to combine the equations (14) – (16) as a matrix equation

$$\begin{pmatrix} \Delta x_{n+1} \\ \Delta \xi_{n+1} \\ \Delta p_{n+1} \end{pmatrix} = \mathbf{G} \begin{pmatrix} \Delta x_n \\ \Delta \xi_n \\ \Delta p_n \end{pmatrix} + \mathbf{H} \Delta v_n \quad (17)$$

Where $\mathbf{G}$ is a 3×3 matrix and $\mathbf{H}$ is 3×1 , the details are also shown in the appendix.

With the form of (17) it is clear that the control input is $u = \Delta v_n$, and we can use standard linear control system analysis method to deal with it.

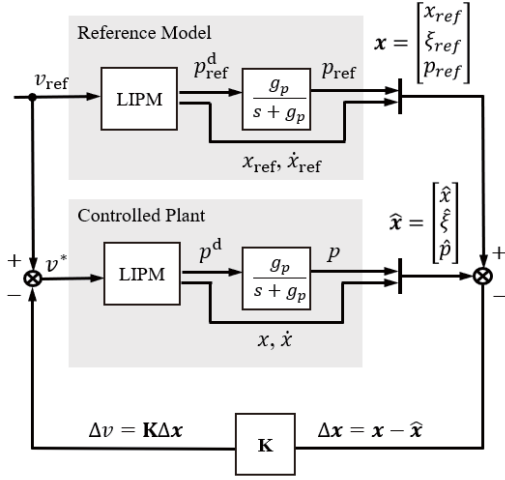


Fig. 3. LIPM CoM-CP-ZMP controller

C. Controllability proof and pole assignment

The controllability proof is to calculate the rank of controllability matrix

$$W_c = (H \ G \ H \ G^2 \ H) \quad (18)$$

If $\text{rank}(W_c) = 3$, this system is complete controllable. Afterwards we replace the parameters to the corresponding values about the actual application and prove that it is always controllable.

Therefore, we can use the state feedback controller

$$u = \Delta v_n = K \Delta x_n \quad (19)$$

$$K \equiv (k_1 \ k_2 \ k_3)$$

Fig.3 shows the block diagram of the proposed controller. The reference model is just a generator to provide the reference state $\mathbf{x} = (x_{ref}, \xi_{ref}, p_{ref})$. When the state of controlled plant $\hat{\mathbf{x}} = (\hat{x}, \hat{\xi}, \hat{p})$ has some deviations $\Delta \mathbf{x} = \mathbf{x} - \hat{\mathbf{x}}$, the pivot of the controlled LIPM is changed to $v^* = v_{ref} - \Delta v = v_{ref} - K \Delta \mathbf{x}$ for controlling it to no deviation, that is, $\Delta \mathbf{x} = (0,0,0)$.

We design the controller on pole assignment. The characteristic equation of close-loop system is

$$|zI - G + HK| = 0 \quad (20)$$

According to the root of this equation is corresponding the assigned poles, we can get the solution value of the feedback gain K . For the convergence, the three poles of close-loop system should be inside of the unit circle on the complex plane.

III. APPLICATION TO THE ACTUAL ROBOT

In Section II we proposed the control method just used for the LIPM, but there are also some details when applying this method to the actual robot. These details will be presented in this Section. Fig.4 shows the changed block diagram for robot, and we will focus on the differences from the Fig.3.

A. Hardware platform WALKER

The target robot for our control is WALKER shown in Fig.1. It is a humanoid robot of 50.7kg weight and 12 degrees of freedom (DOF) in the leg. In the arms there are another DOFs but we don't use them in our experiments. The robot body is equipped with an Intel NUC with core i7 3.1GHz. We use the software Beckoff TwinCAT in Windows to realize a real-time system and EtherCAT to make the communication between NUC and the servo joints.

B. LIPM to robot

We can find the significant difference between Fig.3 and Fig.4 is that the controlled plant is our robot instead of LIPM. But as robot with position-controlled joints, the inputs must be the angle of each joint, so it is necessary to interpret the process from the output of controller Δv to the inputs of robot.

As shown in Fig.4, there is another LIPM controlled by $v^* = v_{ref} + \Delta v$, then it will generate the trajectory of CoM (also we can get the velocity of CoM). Because of the dynamics of LIPM as (2), it just like v^* corresponding the acceleration of CoM $\ddot{x}$ and we can integrate $\ddot{x}$ to obtain $\dot{x}$ and x . Then we use x and $\dot{x}$ to generate the joint angles trajectory of robot as shown in Fig.5.

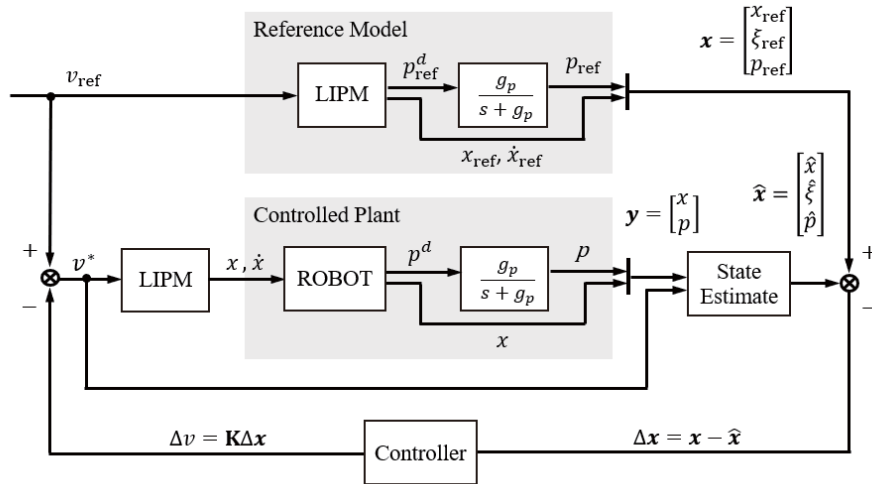


Fig. 4. Control structure on the robot

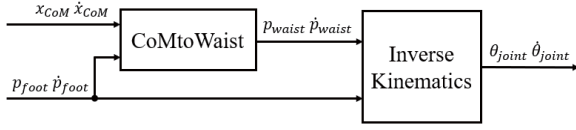
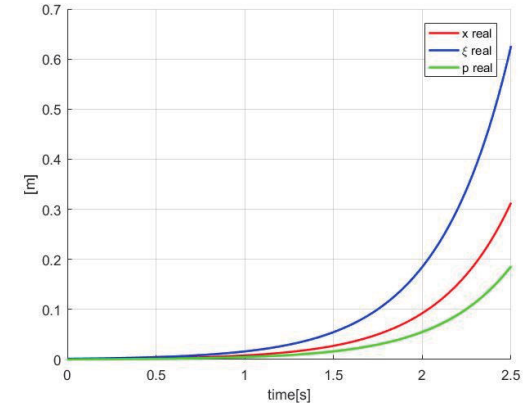


Fig. 5. Process from LIPM to robot

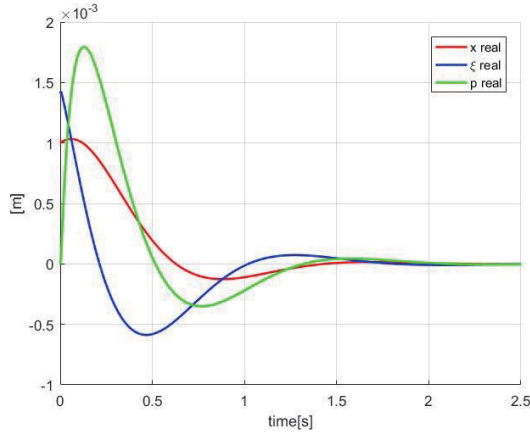
It is an easy way to let the CoM fixed with the waist and if we plan trajectories of both feet, we can use inverse kinematics to solve the angle and velocity of each joint directly. However, the weight of the robot is not all on the upper body, so it is better to consider the weight of two legs. Then there must be a process of ‘CoM to Waist’ as shown in Fig.5. Using planned CoM’s trajectory and both feet’s trajectory to calculate the waist’s trajectory. According to the relative position relationship of waist and both feet, we can obtain θ_{joint} and $\dot{\theta}_{joint}$ by inverse kinematics, and they are the exact inputs of the robot.

C. State Estimation

We can’t observe the three states of the robot directly, instead there is state estimation. By using the servo joint encoders, IMU and six axis force sensors, measurement and calculation of the CoM’s position and ZMP is possible. And we use the Kalman filter to estimate the states we need.



(a) Initial state has deviations without control



(b) Initial state has deviations with control

Fig. 6. Simulation results

IV. SIMULATION AND EXPERIMENTS

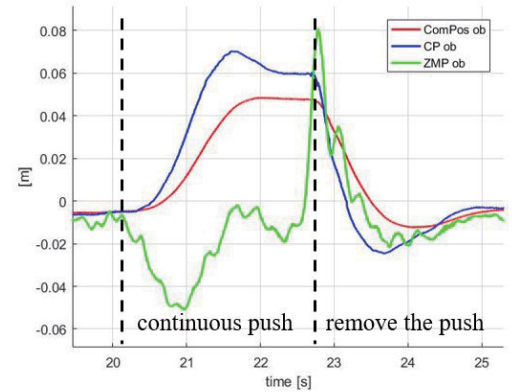
A. Simulation

To confirm the validity of the proposed control method, the simulation results in MATLAB Simulink are shown in Fig. The block diagram structure is just like Fig.3, and there are some deviations of the controlled LIPM. The reference state of LIPM is $x = 0$, $\xi = 0$ and $p = 0$. The reference VSP is also $v = 0$. When the initial state of LIPM is not $(0,0,0)$, it will be divergent according to its dynamics as shown in Fig.6(a). But the controller can make it converge to the reference state as shown in Fig.6(b).

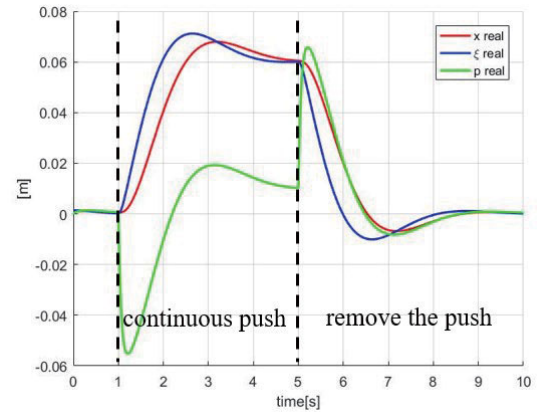
It is a simple but effective simulation verification, and when we change the pole of close loop, it just has influence on the speed and characteristics of convergence. Therefore, the pole assignment on the actual robot should be based on the characteristics of experiments.

B. Experiments

Our experiment platform is UBTECH’s WALKER, a biped humanoid robot with position controlled joint. The implementation method has been shown in Section III. We take the experiments on standing push first. Because of the divergent of LIPM, it is also significant to hold on standing



(a) Experiment result of continuous push and removed



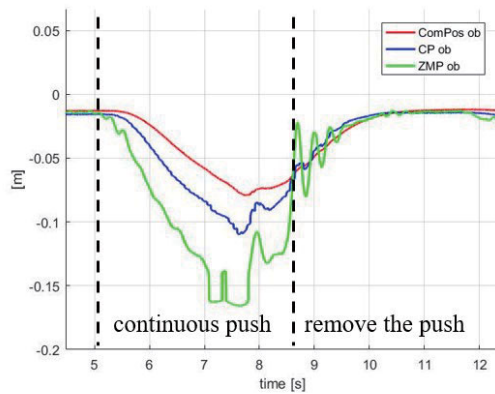
(b) Simulation result of continuous push and removed

Fig. 7. ‘Resistant reaction’ when pushed

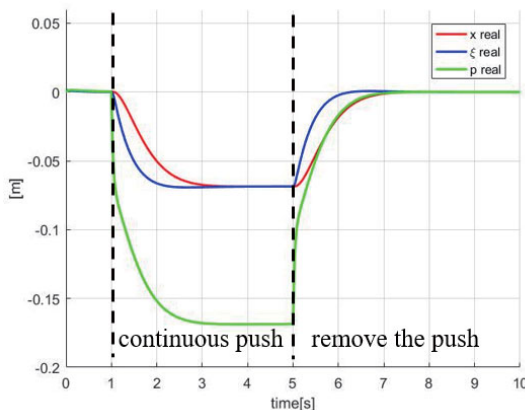
without any big disturbance. But more importantly, it shows a reaction when imposed continuous push as shown in the Video. And when we remove the push it will recover to the state of standing.

The observed state, CoM, CP, and ZMP are shown in Fig.7(a). For clear analysis only the lateral plane is controlled and we push the robot from the left or the right side. From about 20 second to 23 second in this figure, the CoM trajectory is changed to resist the continuous push. It is worth noting that the direction of push is from left to right and just contrary to the direction of CoM's movement. Then we could see the 'resistant' of the continuous push in the video. And when the push is removed it will be controlled to the normal state of standing. Therefore, we have reasons to believe that this control method is effective to deal with some disturbance and control the state to the reference.

To verify the experiment result, we also find this performance of 'resistant' in the simulation as shown in Fig.7(b). The continuous push is model as a constant disturbance of ZMP due to the fact that the ZMP will be changed obviously if pushed. From 1 second to 5 second in this figure there is a constant disturbance of ZMP and after 5 second it is removed. And we find the similar performance with the experiment result, both the 'resistant' when continuous disturbance and converge when the disturbance removed. This



(a) Experiment result of continuous push and removed



(b) Simulation result of continuous push and removed
Fig. 8. 'Compliant reaction' when pushed



Fig. 9. 'Compliant reaction'

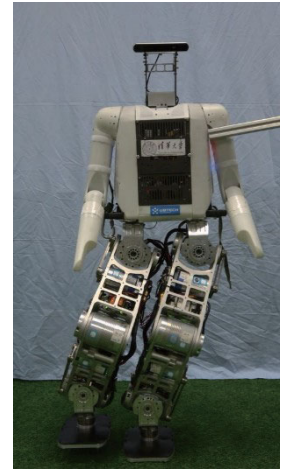


Fig. 10. Push recovery on one foot

experiment result and simulation appear in the same pole assignment.

In addition, we find another performance with the continuous push in another pole assignment as shown in Fig.8(a), Fig.9 and in the video. Instead of 'resistant', it is 'compliant' with the push. The direction of CoM's movement is from left to right and the same as the push. We could see the change of CoM, CP and ZMP about continuous push from about 5 second to 9 second in this figure. And it will also be converged to the reference state (0,0,0) when the push is removed. We also find the same performance in the simulation as shown in Fig.8(b).

Compared these two kinds of pole assignment, the difference point is performances about continuous push and the common point is convergence to the reference state when the push is removed. In general, it shows the ability of standing push recovery without changing the placement of foot.

The robot also performed standing push recovery on only one foot as shown in Fig.10. The changing of CoM also takes effect when pushed although the stable margin of ZMP is too small. But if the robot is already rotating about the edge of the foot, it is unable to be controlled to the reference state.

V. CONCLUSION

In this paper, we proposed a control method based on the LIPM dynamics to control the CoM, CP, and ZMP synchronously. When any of the three states has some disturbance, we expected the controller can eliminate these deviations and control to the reference state. Considering the digital system of the robot, the theory analysis was based on discrete control system and we designed the controller on pole assignment of discrete system. The validity of the controller was verified through simulation and experiments. As the target robot, we used a humanoid biped robot with 12 DOF in the leg, WALKER, which is position-controlled of each joint. Using our method, the robot presented the ability of standing push recovery with double foot or single foot supported. We found two kinds of performance with the continuous push: 'resistant reaction' and 'compliant reaction' on different pole assignments. And both of them showed the convergence to the reference state without external disturbance.

It is also possible that using this method in biped walking. As we said in Section IV before, the rotation of the foot about any edge will cause an abnormal measurement of ZMP. Therefore, the improvement of the proposed control method can be expected by applying some adjustment of the foot. And it will control to the reference state of the LIPM trajectory used for walking. That is our future work.

APPENDIX

We don't give the specifics of equations (11) – (17) in Section II due to the length of them, so we will present these now. According to (6) – (10), the state transition equation from (x_n, ξ_n, p_n) to $(x_{n+1}, \xi_{n+1}, p_{n+1})$ is

$$x_{n+1} = \left[\cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] x_n + \left[\sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] \xi_n + \left[1 - \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] v_n \quad (21)$$

$$\xi_{n+1} = \left[\cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] \xi_n + \left[1 - \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] v_n \quad (22)$$

$$p_{n+1} = \left\{ \frac{g_p(\alpha-1)}{\alpha g_p^2 - \omega^2} \left(\frac{\omega}{\sqrt{\alpha}} + g_p \right) \left[\cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - e^{-g_p T} \right] \right. \\ \left. + \left[\frac{g_p(\alpha-1)}{\alpha g_p^2 - \omega^2} \left[\frac{\omega}{\sqrt{\alpha}} \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + g_p \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \frac{\omega}{\sqrt{\alpha}} e^{-g_p T} \right] \right] \right\} \xi_n \\ + \left\{ 1 + \frac{g_p(\alpha-1) \left[g_p \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \frac{\omega}{\sqrt{\alpha}} \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] + e^{-g_p T} (\omega^2 - g_p^2)}{\alpha g_p^2 - \omega^2} \right\} v_n \quad (23)$$

To simplify the final representation, let

$$p_n = P_A x_n + P_B \xi_n + P_C v_n \quad (24)$$

$$P_A = \frac{g_p(\alpha-1)}{\alpha g_p^2 - \omega^2} \left(\frac{\omega}{\sqrt{\alpha}} + g_p \right) \left[\cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - e^{-g_p T} \right] \quad (25)$$

$$P_B = \frac{g_p(\alpha-1)}{\alpha g_p^2 - \omega^2} \left[\frac{\omega}{\sqrt{\alpha}} \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + g_p \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \frac{\omega}{\sqrt{\alpha}} e^{-g_p T} \right] \quad (26)$$

$$P_C = 1 + \frac{g_p(\alpha-1) \left[g_p \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \frac{\omega}{\sqrt{\alpha}} \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \right] + e^{-g_p T} (\omega^2 - g_p^2)}{\alpha g_p^2 - \omega^2} \quad (27)$$

It can be found that P_A , P_B and P_C are not related to x_n , ξ_n and v_n , so it is easy to calculate the partial derivative of f_3 in (16) (the same with (14) and (15)). Then we obtain for the equation (17)

$$G = \begin{pmatrix} \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) & \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) & 0 \\ 0 & \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) + \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) & 0 \\ P_A & P_B & 0 \end{pmatrix} \quad (28)$$

$$H = \begin{pmatrix} 1 - \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \\ 1 - \cosh\left(\frac{\omega}{\sqrt{\alpha}}T\right) - \sinh\left(\frac{\omega}{\sqrt{\alpha}}T\right) \\ P_C \end{pmatrix} \quad (29)$$

When we decide the four parameters, ω , α , T , and g_p , it is possible to prove the controllability and design the controller by pole assignment.

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